Putting Small Shapes Together to Make Bigger Shapes Answer Key

Kindergarten / Grade 1

Quick Answers

1. 8	4. —	7. —	10. 2, 3, 6
2. 6	5. 12	8. 12	
3. —	6. —	9. —	

Curriculum

Level: Kindergarten / Grade 1
K.G / 1.G / 2.G / 3.G.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

5. *If they got 7: added the two numbers instead of counting three triangles in each of the four trapezoids*
8. *If they got 6: stopped after one hexagon and forgot to double for the second hexagon*

How to read this key. Three different questions are mixed on the sheet on purpose, so the child has to decide what a problem is asking before answering. *How many pieces* questions are counted group by group. *Draw and name* questions are checked by looking at the picture: the pieces must touch along whole sides with no gaps and no overlaps. *Which uses more* questions are answered by comparing two counts, never by comparing how big one piece looks.

1. A rectangle covered by square tiles: 2 rows of 4.

Each row is one group of the same size, so count by rows rather than one tile at a time. There are 2 rows, and each row holds 4 tiles:

$$4 + 4 = 8 \qquad \text{or} \qquad 2 \times 4 = 8$$

The rectangle is covered by $\boxed{8}$ square tiles.
Ask the child: "How do you know you did not count a tile twice?" Counting row by row is the answer.

2. A hexagon covered by rhombuses, and each rhombus by triangles.

This is a two-layer cover. Work from the small piece up: each of the 3 rhombuses is made of 2 green triangles, so there are 3 equal groups of 2.

$$2 + 2 + 2 = 6 \qquad \text{or} \qquad 3 \times 2 = 6$$

The hexagon takes $\boxed{6}$ green triangles.
This is the fact worth remembering: six green triangles fill one yellow hexagon.

3. Two identical squares, one whole side touching.

Slide the two squares together along a full side. The two touching sides disappear inside the build, and the outside edge is now four straight sides with four square corners — but one pair of sides is twice as long as the other.
The new shape is a $\boxed{\text{rectangle}}$ (twice as long as it is tall).
Check the drawing, not just the word: the squares must share a whole side. If a child slides one square only halfway along the other, the outline is not a rectangle, and that is a useful thing to notice out loud.

4. Ben's triangles versus Mia's rhombuses.

Both children covered the same hexagon, so the question is only about *how many pieces* each one used — not about how big a piece is.
Ben used 6 pieces. Mia used 3 pieces. Six is more than three, so Ben used more pieces: $\boxed{6 > 3}$.
Watch for: answering "Mia, because her pieces are bigger." Bigger pieces cover the same hexagon with *fewer* of them — the opposite of more.

5. Four trapezoids, each made of three triangles.

Four equal groups of three. Count the groups, then count by threes:

$$3 + 3 + 3 + 3 = 12 \qquad \text{or} \qquad 3 \times 4 = 12$$

Nina uses $\boxed{12}$ green triangles.
Watch for: the answer 7, which comes from adding the two numbers in the problem instead of making four groups of three.

6. Two identical right triangles, longest sides touching.

The longest side of a right triangle is the side across from the square corner. When the two longest sides are laid exactly on top of one another, the two square corners land on opposite ends and the outline closes into four sides with four square corners.
The shape is a $\boxed{\text{rectangle}}$.
For the second drawing, any other correct build counts. The two most common are:

- a *larger triangle*, made by joining the two shorter legs of equal length;
- a *parallelogram*, made by joining two equal sides but flipping one triangle so the corners point the same way.

The point of part two is that the same two pieces make more than one shape, so the pieces alone do not decide the answer — how they are joined does.

7. Eight square tiles or sixteen triangle tiles.

Same rectangle, two ways to cover it. Compare the two counts:
8 square tiles against 16 triangle tiles. Eight is less than sixteen, so the square-tile covering uses fewer pieces: $\boxed{8 < 16}$.
Why it comes out that way: it takes two triangle tiles to cover one square tile, so the triangle covering always needs twice as many pieces. Smaller pieces always means more of them.

8. Two hexagons, each built from trapezoids, each trapezoid from triangles.

Three layers, so build up one layer at a time.
One hexagon holds 2 red trapezoids, and each trapezoid holds 3 green triangles, so one hexagon takes $2 \times 3 = 6$ green triangles. Ari builds 2 hexagons the same way, so double that count:

$$6 + 6 = 12 \qquad \text{or} \qquad 2 \times 3 \times 2 = 12$$

Ari uses $\boxed{12}$ green triangles.
Watch for: the answer 6, which is one hexagon only — the second hexagon was forgotten.

9. Four green triangles making one larger triangle.

The standard build: put one triangle point-up, put a second and a third beside it also point-up in a bottom row of three, then turn the fourth triangle point-down and drop it into the gap between the first two. The result is one big triangle whose side is twice as long as a small triangle's side.
A correct explanation says something like: *"The new shape has three straight sides and three corners, so it is a triangle. Each big side is two small sides in a row."*
Accept any wording that names three straight sides and three corners. This is an open build — $\boxed{\text{open response, check the drawing and the reasoning}}$.
Common wrong build: four triangles all pointing the same way, which leaves gaps and is not a covering at all.

10. Challenge: order the three coverings.

Read the count for each way of covering the one hexagon:

- red trapezoids: 2 pieces
- blue rhombuses: 3 pieces
- green triangles: 6 pieces

Now put the counts in order from fewest to most: $\boxed{2, 3, 6}$ — trapezoids, then rhombuses, then triangles.
The big idea: the pieces get smaller as you go down that list, and the count gets bigger. Covering one shape always takes more pieces when the pieces are smaller.